

Shrishti Tomar

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#)

Summary

Computer Science undergraduate (B.Tech, 2027) who builds and deploys full-stack, distributed web applications and AI-powered engines with Next.js, React, Node.js, Express, Redis, and PostgreSQL. Built production projects leveraging Retrieval-Augmented Generation (RAG), asynchronous job queue processors, WebSockets, multi-container Docker architectures, and AWS EC2 deployments.

Education

Bhagwan Parshuram Institute of Technology (Guru Gobind Singh Indraprastha University)

Delhi, India

B.Tech, Computer Science and Engineering; CGPA: 8.29 / 10

Expected June 2027

- Coursework: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, OOP, System Design

Technical Skills

Languages: C++, JavaScript, TypeScript, Python, SQL, HTML5, CSS3

Frontend: React 19, Next.js (App Router, SSR), Vite, Redux Toolkit, TanStack Query, Tailwind CSS, Recharts

Backend: Node.js, Express.js, REST APIs, BullMQ, WebSockets (Socket.IO), JWT Authentication, Webhooks

Databases: PostgreSQL, MongoDB Atlas, Redis, Supabase, Prisma ORM, MongoDB Aggregation

AI and LLM: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Vector Embeddings, Gemini API, Agentic Workflows

Cloud and DevOps: AWS EC2, Docker, Docker Compose, Vercel, Git, GitHub, Linux, Postman, CI/CD Pipelines

Projects

RepoInsight | *AI Code Search Platform — React, Vite, LangGraph, Express, MongoDB, AWS*

[GitHub](#) | [Live](#)

- Built a Retrieval-Augmented Generation platform that clones public GitHub repositories and indexes source files as vector embeddings, enabling natural-language search across codebases of up to 10,000+ files.
- Designed a LangGraph multi-agent layer routing queries across code search, dependency mapping, and summarization agents, cutting average answer latency from 12s to 3s.
- Containerized with a multi-stage Docker build packaging the Vite frontend and Express backend into a single container (size down 85%) and deployed to AWS EC2, with JWT auth and chat history persisted in MongoDB Atlas.

TaskForge | *Distributed Task Queue Engine — Next.js, Express, BullMQ, Redis, Socket.IO, Docker*

[GitHub](#) | [Live](#)

- Engineered an asynchronous task processing engine using Express, BullMQ, and Redis to offload CPU-intensive operations (OCR extraction, PDF generation, scraping) across 5 concurrent worker threads with sub-1.8ms Redis transport latency.
- Implemented a Socket.IO WebSocket telemetry engine broadcasting live job progress (0% to 100%) and audit logs directly to client dashboards, eliminating HTTP polling overhead by 100%.
- Orchestrated a 5-container microservices stack via Docker Compose, modeled PostgreSQL schemas using Prisma ORM with exponential backoff retries (max 3 attempts), and implemented dual-token JWT authentication with RBAC.

DropAlert | *E-Commerce Price Tracker — Next.js, Supabase, Firecrawl, Redis, Resend*

[GitHub](#) | [Live](#)

- Built a price tracker that extracts structured product data from e-commerce listings via the Firecrawl API, bypassing anti-bot protections that block conventional scrapers.
- Automated periodic price checks with cron jobs and Redis caching across 50+ tracked products, comparing results against user-defined targets.
- Configured templated transactional email alerts via Resend, notifying users within 5 minutes of a price drop, and charted price history.

Achievements

- **GTBIT Hackathon:** Secured 3rd place among 80+ competing teams.
- **HackMait Hackathon:** Placed in the top 17 teams out of 700+ participants.
- **Competitive Programming:** Solved 400+ Data Structures and Algorithms problems on LeetCode and other platforms.

Training and Leadership

- Junior Counselling Member, Namespace Community
- Web Development Training, Winnovation
- Data Structures and Algorithms Certification, Apna College